



Multi-camera view Face recognition

Build an End to End Image Classification/Recognition
Application



Scope

Overview

Image classification

Deliverables

Face recognition

Daily reporting

Vision

Deep Learning Algorithms

GUI

Analysis

Optimization

Process



Overview

To build our face recognition system, we'll first perform face detection, extract face embeddings from each face using deep learning, train a face recognition model on the embeddings, and then finally recognize faces in both images and video streams with OpenCV.





OpenCV Algorithms

As we know, OpenCV comes equipped with three face recognizers :

1

EigenFace Recognizer: This can be created with
`cv2.face.createEigenFaceRecognizer()`

2

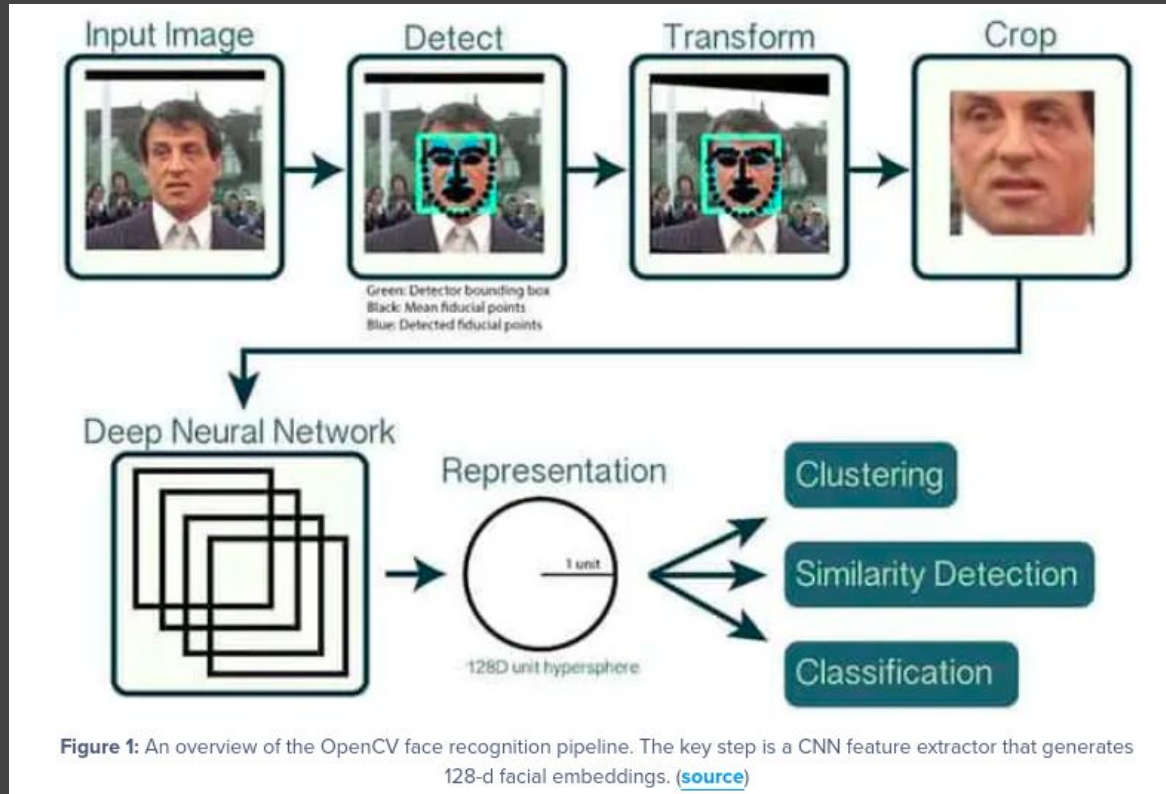
FisherFace Recognizer: This can be created with
`cv2.face.createFisherFaceRecognizer()`

3

Local Binary Patterns Histogram (LBPH): This can be created with
`cv2.face.LBPHFisherFaceRecognizer()`



How OpenCV's face recognition works





Demo

Face Detection/Recognition Algorithm

Camera Network vs single Camera

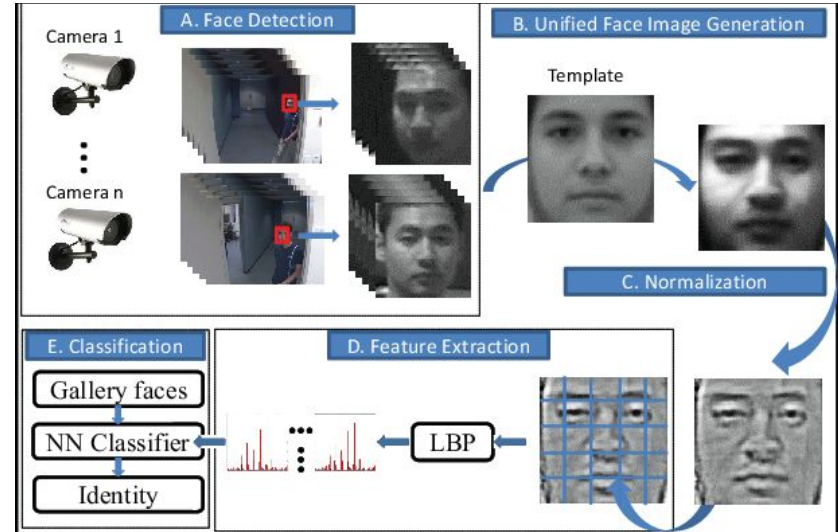
With camera networks, we can capture multi-view videos which allow us to further integrate information spatially across view **angles**. In that case, one usually needs to model the **dynamics of pose changes** in the training phase and **estimate pose** in the testing phase.



Fig 1 : Images acquired by a multi-camera network. Each **column corresponds to a different camera**, and each **row corresponds to a different time instant and subject**. Note that, under unconstrained acquisition, it is entirely possible that none of the images are frontal in spite of using five cameras to observe the subject [32]

Conditions 1 : Collaboration

The method should be able to collaboratively utilize information collected by multiple cameras and arrive at a **multi-view representation** from it, as opposed to perform recognition for each view individually and then fusing the result.



Conditions 2 : Pose variation

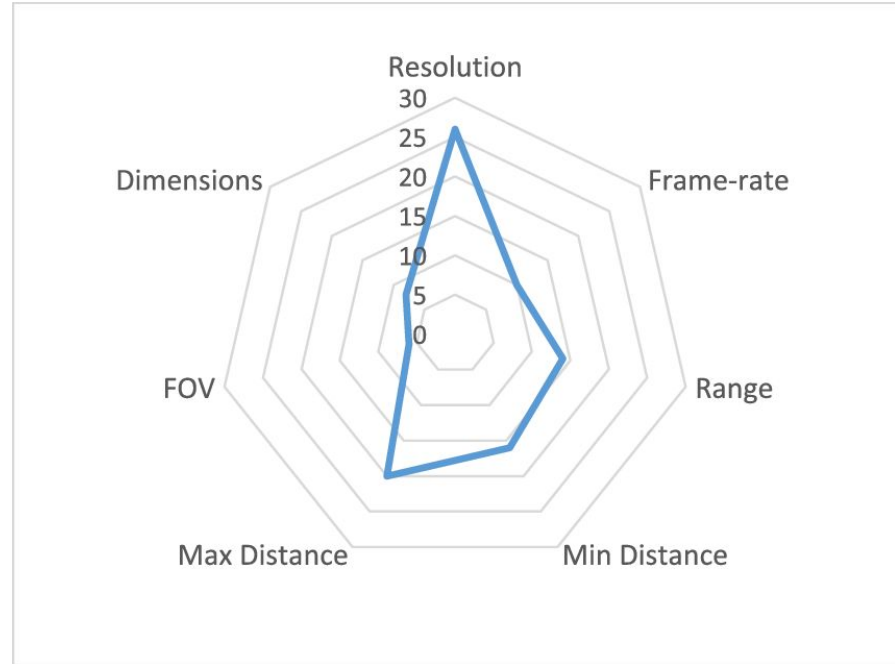
The method should be able to tackle **pose variations** effectively, as this is the major concern of a **multi-view face recognition system**.

Figure 1

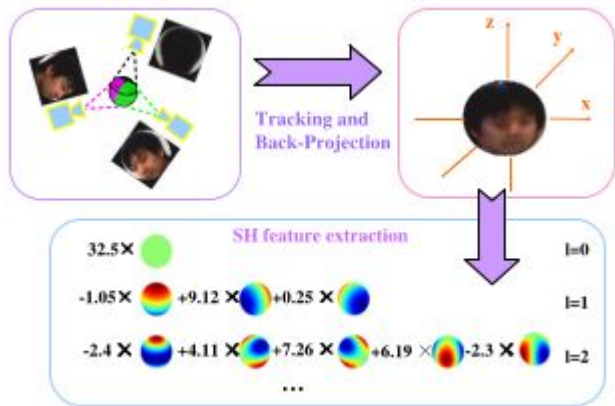


Conditions 3 : Real situation

The method should work on data whose acquisition conditions are as close to practical surveillance situations as possible. These conditions include: reasonable **distance** between subject and cameras, relatively **low resolution** in the face region, uncontrolled **pose variations**, uncontrolled subject **motion**, and possible **interruptions** in acquisition (say, the subject moves out of the field of view of a camera) etc.



Face Tracking from Multi-view Videos



Spherical Harmonics : Feature extraction. We first obtain the **texture map** of the human head on the surface of a spherical model through **back projection of multi-view images** captured by the camera network, then represent it with spherical harmonics.



Pose-Free Feature Based on Spherical Harmonics

In the vision community, following the work of Basri and Jacobs [5], researchers have used **SH** to understand the impact of **illumination variations** in face recognition [30, 39].



Fig 2 : Sample tracking results for a multi-view video sequence. 5 views are shown here. Each **row** of images is captured by the same **camera**. Each **column** of images corresponds to the same **time-instant**.

- Lambertian Reflectance and Linear Subspaces

Ronen Basri, *Member, IEEE*, and David W. Jacobs, *Member, IEEE*

Abstract—We prove that the set of all Lambertian reflectance functions (the mapping from surface normals to intensities) obtained with arbitrary distant light sources lies close to a 9D linear subspace. This implies that, in general, the set of images of a convex Lambertian object obtained under a wide variety of lighting conditions can be approximated accurately by a low-dimensional linear subspace, explaining prior empirical results. We also provide a simple analytic characterization of this linear space. We obtain these results by representing lighting using spherical harmonics and describing the effects of Lambertian materials as the analog of a convolution. These results allow us to construct algorithms for object recognition based on linear methods as well as algorithms that use convex optimization to enforce nonnegative lighting functions. We also show a simple way to enforce nonnegative lighting when the images of an object lie near a 4D linear space. We apply these algorithms to perform face recognition by finding the 3D model that best matches a 2D query image.

Index Terms—Face recognition, illumination, Lambertian, linear subspaces, object recognition, specular, spherical harmonics.





Face Recognition from a Single Training Image under Arbitrary Unknown Lighting Using Spherical Harmonics

Lei Zhang, *Student Member, IEEE*, and Dimitris Samaras, *Member, IEEE*

Abstract—In this paper, we propose two novel methods for face recognition under arbitrary unknown lighting by using spherical harmonics illumination representation, which require only one training image per subject and no 3D shape information. Our methods are based on the recent result which demonstrated that the set of images of a convex Lambertian object obtained under a wide variety of lighting conditions can be approximated accurately by a low-dimensional linear subspace. We provide two methods to estimate the spherical harmonic basis images spanning this space from just one image. Our first method builds the statistical model based on a collection of 2D basis images. We demonstrate that, by using the learned statistics, we can estimate the spherical harmonic basis images from just one image taken under arbitrary illumination conditions if there is no pose variation. Compared to the first method, the second method builds the statistical models directly in 3D spaces by combining the spherical harmonic illumination representation and a 3D morphable model of human faces to recover basis images from images across both poses and illuminations. After estimating the basis images, we use the same recognition scheme for both methods: we recognize the face for which there exists a weighted combination of basis images that is the closest to the test face image. We provide a series of experiments that achieve high recognition rates, under a wide range of illumination conditions, including multiple sources of illumination. Our methods achieve comparable levels of accuracy with methods that have much more onerous training data requirements. Comparison of the two methods is also provided.

Index Terms—Face recognition, spherical harmonics illumination representation, 3D face morphable models, illumination invariance.





Measure Ensemble Similarity

1

Given two multi-view video sequences with m and n frames (Every “frame” is actually a group of images, each captured by a camera in the network.), respectively, we generate 2 ensembles of feature vectors.

2

To achieve video-level recognition, we are interested in measuring the similarity between these two sets of vectors. Now, we calculate the ensemble similarity as the limiting Bhattacharyya distance in RKHS following [41].

3

In experiments, we measure the ensemble similarity between feature vectors of a probe video and those of all the gallery videos.

4

The gallery video with the shortest distance to the probe is considered as the best match.

Statistics : Power BI

Integrating statistics in one web application platform.

- 01 | hourly
- 02 | Daily
- 03 | By Classroom Subject
- 04 | By Seasons/weather/holidays
- 05 | Other ...





Thank you.

